

# DARREL DANADYAKSA POLI

Software Engineer | AI Engineer | AI Researcher

Depok, Indonesia | darrel.danadyaksa19@gmail.com | linkedin.com/in/darrel-danadyaksa-poli

Final Year Computer Science student with strong expertise across Software Engineering, Backend Architecture, and Artificial Intelligence. Experienced in AI research at **NVIDIA**, production speech model distillation at Meeting.ai, building scalable web and mobile applications (Django, Go, Laravel, Flutter), and leading software engineering teams.

## EDUCATION

### University of Indonesia

Undergraduate Computer Science

Depok, Indonesia

Aug 2022 – Aug 2026

- **Focus Areas:** Software Engineering, Web & Mobile Development, Deep Learning, Natural Language Processing, Computer Vision, Algorithms & Complexity.
- **Academic Mentorship:** Teaching Assistant for Software Engineering Project, Operating Systems, Calculus 1 (Regular & International), and Calculus 2.
- **Student Psychological Support:** Peer Counselor at Curhat Sama Panda, supporting student mental health at Faculty of Computer Science.

## WORK EXPERIENCE

### NVIDIA

AI Research Intern

Remote

Jan 2025 – Jun 2025

- **Model Expressivity Benchmarking:** Evaluated Non-Transformer sequence model architectures (**RWKV**) against standard **Transformers** for formal regular language recognition tasks, delivering comparative insights to senior research supervisors.
- **Sequence Architecture Analysis:** Investigated structural representations and formal language capacity across **Transformer**, **RNN**, **LSTM**, and **GRU** models.
- **Research Dissemination:** Conducted monthly technical presentations detailing empirical findings, loss convergence dynamics, and theoretical trade-offs to guide architectural research directions.

### Meeting.ai

AI Engineer Intern

Jakarta, Indonesia

Jul 2025 – Dec 2025

- **ASR Performance Stress-Testing:** Benchmarked Automatic Speech Recognition (ASR) pipelines under high concurrency (10+ simultaneous sessions) to isolate latency bottlenecks and ensure deployment stability.
- **Speaker Verification Optimization:** Compression-optimized a Speaker Verification model using knowledge distillation, significantly reducing parameter footprint while maintaining accuracy and achieving faster inference throughput.
- **Active Learning Pipeline:** Architected a user-in-the-loop correction mechanism within the Speaker Verification pipeline to dynamically resolve clustering ambiguities and continuously improve downstream model accuracy.

### Intergalactic Science Kingdom

AI Engineer & Back End Engineer (Volunteer)

Remote

Jan 2026 – Jul 2026

- **Marie Chan Full-Stack Architecture:** Architected and deployed *Marie Chan*, an interactive AI system combining Go (Golang) and Django backends for real-time API performance and backend stability.
- **Agentic AI & BDD/TDD Workflows:** Applied Behavior-Driven Development (BDD) and Test-Driven Development (TDD) to guide and validate software engineering workflows powered by Agentic AI.
- **LLM Pipeline Orchestration:** Integrated OpenRouter API for dynamic multi-model LLM routing, optimizing inference latency, context window management, and speech synthesis pipelines.

### Indonesia AI x OMDENA Forest Fire Project

AI & Data Science Contributor

Indonesia

Aug 2022 – Sep 2022

- **Data Engineering:** Curated, preprocessed, and normalized multi-source environmental datasets to enhance training data quality for forest fire risk prediction models.
- **Model Fine-Tuning:** Fine-tuned predictive algorithms to maximize early detection accuracy and support proactive community safety initiatives.

## VOLUNTEERING & LEADERSHIP EXPERIENCE

---

### Candidate College

Back End Developer

Indonesia

Jun 2024 – Oct 2024

- **RESTful API & Database Engineering:** Developed and maintained server-side logic, databases, and RESTful APIs using Laravel, collaborating with front-end teams and stakeholders to ensure high performance.
- **Database Modeling & Architecture:** Created Entity-Relationship (EER) diagrams and database models aligned with business needs, enforcing clean code practices, design patterns, and scalable software architecture principles.
- **API Documentation & Leadership:** Led Swagger integration for automated API documentation as Co-Head of the Website Development Division, improving API accessibility for internal and external teams.
- **DevOps & Testing:** Implemented CI/CD pipelines and robust test suites to ensure code quality, maintainability, and streamlined deployments.

### SIKAP

Software Engineer

Indonesia

Aug 2025 – Nov 2025

- **Cross-Platform Engineering:** Engineered a cross-platform anti-bullying information system using Django for the backend and Flutter for the mobile interface, ensuring secure and anonymous reporting.
- **UAT & User-Centered Design:** Conducted User Acceptance Testing (UAT) across dozens of Indonesian schools, utilizing direct user feedback to refine features and improve system stability following ADDIE and UCD principles.

### Student Executive Board (BEM UI)

Coordinator, Information Technology Division

Depok, Indonesia

Jul 2025 – Jan 2026

- **Engineering Leadership:** Directed and reviewed software development workflows, applying industry-standard software engineering methodologies and architectural best practices.
- **Technical Guidance & Quality Control:** Provided hands-on technical guidance on system architecture, maintainability, and quality control mechanisms for ongoing IT projects across multiple divisions.

### Rentang Jiwa

Peer Counselor

Indonesia

Jan 2024 – Jun 2024

- **Client Counseling:** Offered structured peer counseling and psychoeducation using counseling psychology best practices to build client rapport and facilitate psychological support.

## PROJECTS & ACHIEVEMENTS

---

- **SafetyPin Startup (2025):** Co-founded SafetyPin, a social security startup leveraging technology and intelligent reporting systems to enhance personal safety solutions.
- **Multimodal Emotion Detection (Satria Data 2025):** Engineered an end-to-end emotion detection system combining BERT for textual feature extraction, Whisper for speech representations, and VAD for noise filtering.
- **Illegal Parking CV System (Gemastik 2024):** Designed and trained an object detection pipeline to identify illegal parking violations from surveillance feeds.
- **BERT Clickbait Classification (Gemastik 2023):** Fine-tuned a BERT-based transformer model on Indonesian textual datasets to detect clickbait headlines.
- **Garuda Hacks 4.0 Honorable Mention (2023):** Awarded Honorable Mention for building *Tak Sendiri*, a conversational AI agent for preliminary depression screening.
- **Certified Peer Counselor Training (2023):** Completed certified training in peer counseling and psychological support.

## TECHNICAL SKILLS & EXPERTISE

---

- **Software Engineering & Backend:** RESTful APIs, System Architecture, Database Design (EER), Microservices, Agentic AI Workflows, Behavior-Driven Development (BDD), Test-Driven Development (TDD), CI/CD, DevOps.
- **Programming Languages:** Python, Go (Golang), Java, PHP, Rust, Dart, SQL (PostgreSQL, MySQL).
- **Frameworks & Libraries:** Django, Spring Boot, Laravel, Flutter, PyTorch, TensorFlow, Keras, HuggingFace, OpenCV, Scikit-Learn, Pandas, NumPy.
- **Developer Tools & Cloud:** Docker, Git, GCP, Supabase, Postman, Linux/Unix, OpenRouter LLM APIs.
- **Soft Skills & Leadership:** Software Engineering Leadership, Code Review, Mentoring, Critical Thinking, Active Listening, Agile/Scrum.